AquaTech

Water Source Inverter Heat Pump

Temperature Control System for Swimming Pools







AquaTech





TECHNICAL SPECIFICATIONS

WATER SOURCE INVERTER HEAT PUMP							
MODEL			**WM036WSINV**	**WM050WSINV**	**WM060WSINV**	**WM080WSINV**	**WM120WSINV**
Heating Capacity @30/35°C (POOL) 10/7°C (SOURCE)		kW	2.9 – 8.9	4.3 – 15.2	4.8 – 19.5	8.6 - 20	8.0 - 31
Power input		kW	0.4 – 1.8	0.7 – 3.6	0.8 – 4.7	1.5 – 4.8	1.1 – 7.2
Input Current		A	2.2 – 10.2	1.3 – 6.8	1.5 – 8.9	2.8 – 9.1	2.0 – 13.6
COP			4.8 – 6.7	4.2 – 6.1	4.1 – 6.0	4.1 – 5.7	4.3 – 7.2
Power supply		V/Ph/Hz	220 – 240/50/1	380 – 415/50/3			
Compressor type			Rotary	Scroll			
Compressor Qty			1				
Expansion device			Electronic expansion valve				
Controller type			Micro processor based digital thermostat controller with LCD display				
Source temperature			7°C - 25°C				
Refrigerant			R410a				
Noise level		dB(A)	40	40	40	45	45
Condenser heat exchanger (POOL)	Type		Shell & Coil				
	Connection size	mm	Ø40	Ø40	Ø40	Ø40	Ø50
	Flow	m³/h	1 – 5.4	1.5 – 8.6	1.5 – 10.8	2 – 13.5	3 – 18.5
	Pressure drop	bar	0.05	0.07	0.13	0.14	0.15
Evaporator heat exchanger (SOURCE)	Type		Shell & Coil				
	Connection size	mm	Ø40	Ø40	Ø40	Ø40	Ø50
	Flow	m³/h	0.75 – 4.05	0.86 – 6.45	0.86 – 8.10	1.5 – 10.1	2.25 – 13.8
	Pressure drop	bar	0.21	0.25			0.53
Enclosure			Galvanized steel with powder coating				
Unit dimensions		mm	720 x 630 x 750	850 x 745 x 875	850 x 745 x 875	900 x 745 x 875	1480 x 845 x 995
Net weight		Kg	72	120	120	190	285
Suitable for swimming Pool size		Volume m³	24	29	33	50	67
		Surface Area m²	16	18	22	33	44

***All dimensions subject to change.**

STANDARD FEATURES

MITSUBISHI INVERTER COMPRESSOR

COMPACT CABINET FOR EASY TRANSPORTATION & INSTALLATION

TITANIUM HEAT EXCHANGER

RS485 MODBUS COMMUNICATION

PROGRAMMABLE DIGITAL TEMPERATURE CONTROLLER /INDICATOR

PHASE FAILURE RELAY

LOW-AND-HIGH PRESSURE CUT-OUT

COMPRESSOR OVERLOAD PROTECTION

FLOW SWITCH

AXIAL FANS

ANTI - FREEZE PROTECTION

POWDER COATED GI CASING

OPTIONS

SS304 OR SS316L ENCLOSURE

REMOTE MONITORING

BTU METERING

COP METERING

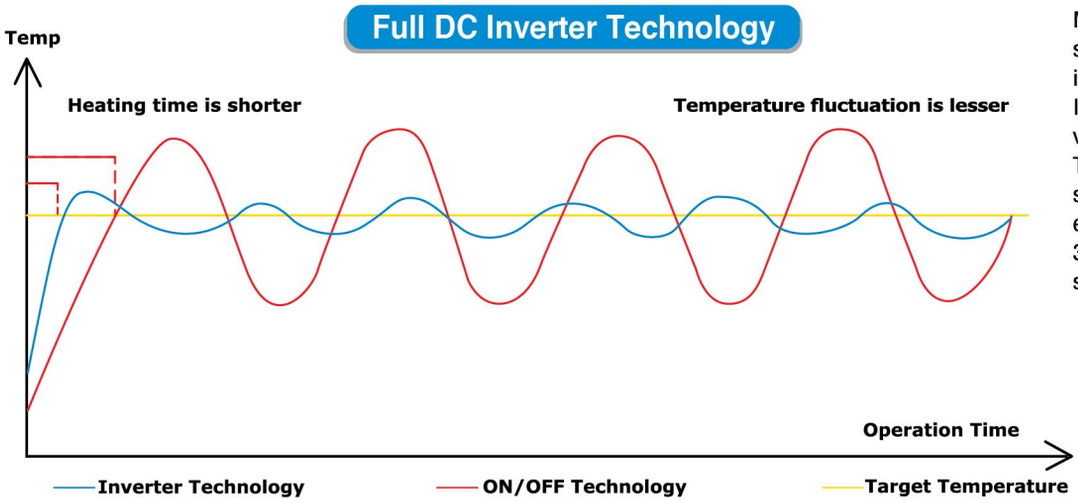






WATER SOURCE INVERTER HEAT PUMP					
WM150WSINV	**WM180WSINV**	**WM240WSINV**	**WM400WSINV**	**WM500WSINV**	**WM600WSINV**
8.6 – 38	14 - 66	17 - 79	23 - 97	28 – 129	34 – 157
1.3 – 9.1	2.1 – 15.4	2.5 - 18	3.5 – 22.3	4.2 – 30.5	5.0 - 36
2.4 – 17.2	3.9 – 29.2	4.7 – 34.1	6.6 – 42.3	7.9 – 57.9	9.4 – 68.3
4.1 – 6.6	4.2 – 6.6	4.3 – 6.8	4.3 – 6.5	4.2 – 6.6	4.3 – 6.6
380-415/50/3					
Scroll					
1					
Electronic expansion valve					
Microprocessor based digital thermostat controller with LCD display					
7°C - 25°C					
R410a					
45	55	55	58	58	58
Shell & Coil					
Ø50	Ø50	Ø80	Ø80	Ø80	Ø80
3 – 22.4	7 - 33	10 – 41	20 – 48	26 – 54	32 – 60
0.17	0.23	0.23	0.30	0.35	0.35
Shell & Coil					
Ø50	Ø50	Ø80	Ø80	Ø80	Ø80
2.25 – 16.8	5.2 – 24.7	7.5 – 30.7	15 - 36	19.5 – 40.5	24 – 45
0.56	0.57	0.60	0.56	0.60	0.72
Galvanized steel with powder coating					
1480 x 845 x 995	2000 x 950 x 2100	2000 x 950 x 2100	2000 x 950 x 2150	2300 x 1100 x 2200	2300 x 1100 x 2300
290	580	650	850	1150	1290
92	120	160	240	350	400
33	80	108	160	240	280

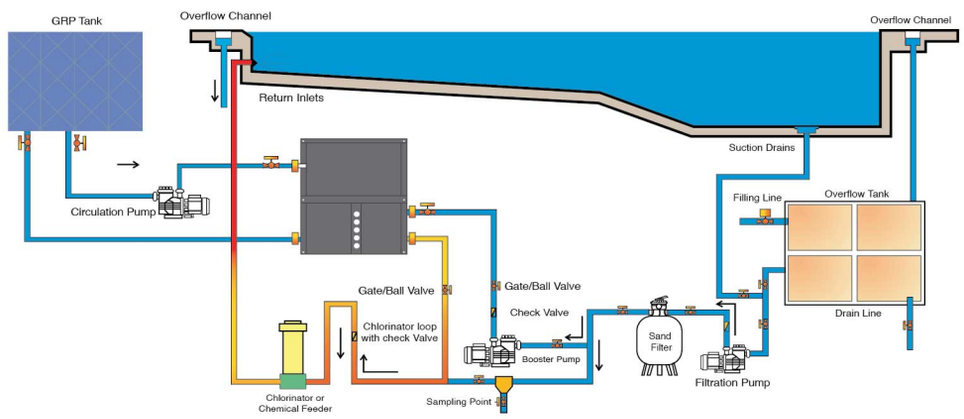
Full DC Inverter Technology



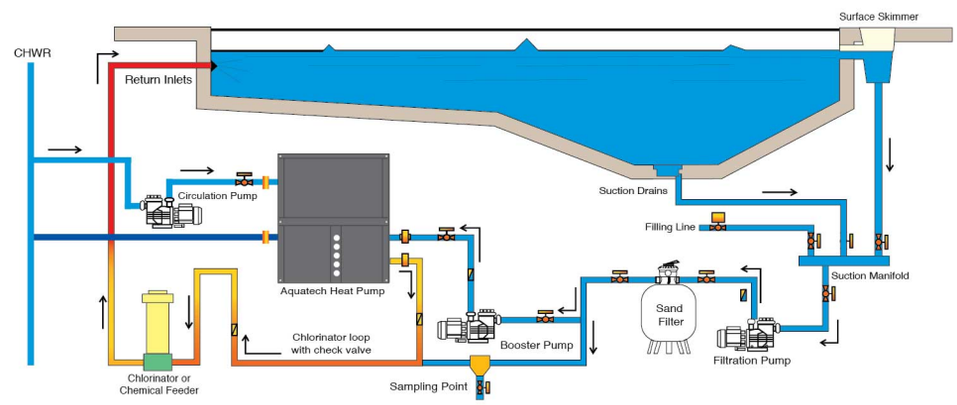
Most heat pumps use single speed compressors. This means it is either off or on. Our DC Inverter Heat Pump uses a variable speed DC compressor. This DC motor runs at the exact speed it needs for optimal efficiency, DC heat pumps average 30-50% more efficiency versus a standard unit.

INSTALLATION SCHEMATIC

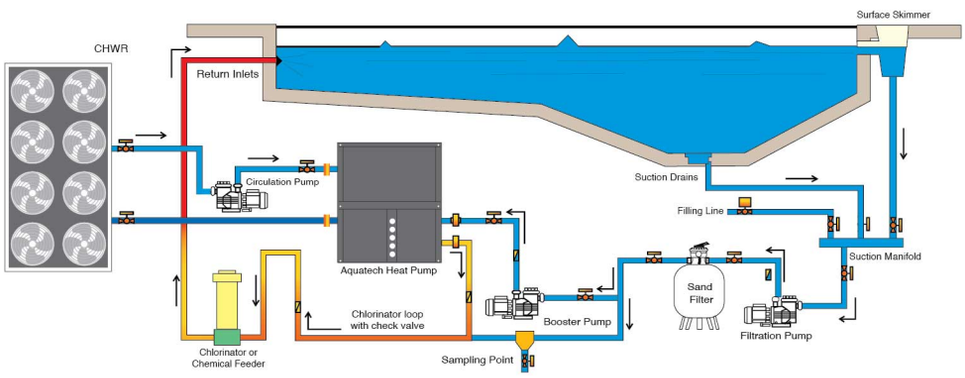
INSTALLATION WITH DOMESTIC WATER AS SOURCE (HEATING ONLY)



INSTALLATION WITH CHILLED WATER RETURN AS SOURCE (HEATING ONLY)



INSTALLATION WITH DRY COOLER (HEATING & COOLING)



WEATHER MAKERS INTERNATIONAL

Weather Makers International Equipment Trading L.L.C

P.O.Box : 20439, Dubai, UAE, Tel: +971 4 3231508
E-mail: info@weathermakersinternational.com
Website: www.weathermakersinternational.com